Validation and AI Use: Tesla (TSLA), Project 1

> Final record for submission (October 19, 2026). The Brightspace PDF `Validation-and-AI-Use.pdf` is a copy of this file.

Decision, version, and as-of boundary

- **Decision and user:** buy-side PM committee. Should the fund initiate a TSLA position, put it on watch/defer, or not initiate?
- **Project of record:** `tsla-proforma-model/` (Streamlit app plus the `tsla_model` engine)
- **Repository commit at freeze:** TODO (Miles): paste the final commit hash
- **Valuation date:** 2026-09-24. **Price:** $378.25 (Yahoo/Nasdaq, as of 2026-09-24). **Risk-free rate:** 5.11% (Treasury 10y CMT, 2026-09-23)
- **Filing cutoff:** 10-K FY2023–FY2025 and 10-Q Q2 2026 (acc. 0001628280-26-049270)
- **Environment:** Python 3.10+, `tsla-proforma-model/requirements.txt`, offline seed data

Data and convention ledger

Material input	Source	As-of	Definition/units	Treatment	Risk
Historical statements	10-Ks FY23–25, 10-Q Q2 2026 (`seed_filings.json`, page refs)	Filing dates	USD M, GAAP	176 tagged values XBRL-verified in live mode	Low
Share price	Yahoo / Nasdaq	2026-09-24	USD/share	Fixed in seed for reproducibility	Moves daily; the value range does not depend on it
Diluted shares	10-Q Q2 2026	2026-06-30	3,540.0M	Unearned 2025 CEO award (423.7M) excluded by default (toggle)	Medium: including them lowers value/share
Cost of equity	CAPM: 5.11% + 1.17 bottom-up beta × 4.5% ERP	2026-09-24	%	Bottom-up beta from 6 peers; observed beta 1.76 shown as an alternative	High: ke is the strongest value driver
Capex	10-Q guidance "in excess of $25B" for 2026	2026	USD M	Later years are judgment	High: drives negative FCFE in 2026–27
Robotaxi revenue	None disclosed	n/a	USD M	Explicit judgment line: $8B by FY2030 in base	High: this is where the market's value sits

Material input	Source	As-of	Definition/units	Treatment	Risk
Model ASPs	Only Model 3/Y vs Other disclosed	FY2025	USD/vehicle	Other models assumed at 2× Model 3/Y price	Medium

Validation register

Claim/output	Failure mode	Test and expected result	Actual	Disposition	Evidence
Engine math is right	Formula error	**Known answer:** proforma.py reproduces the ABG training case of $291.75/share	$291.75	Pass	Lab_09_proforma_engine.md, python proforma.py
Statements articulate	BS doesn't balance; cash doesn't tie	A = L + E, CF cash = BS cash, PP&E/debt/lease/share roll-forwards, every year	All pass in all 4 scenarios	Pass	tsla_model/checks.py, output/visible_output.md
Checks fail loudly	Broken input still produces a value	Break the BS or breach min cash → valuation must refuse	Refuses	Pass	tests/test_valuation_and_checks. ::test_minimum_cash_breach_blo
DCF boundary	g ≥ ke gives nonsense	Terminal g at or above ke → refuse	Refuses	Pass	::test_terminal_growth_at_or_abo
DCF monotonicity	Value moves the wrong way	Higher discount rate → lower value	Lower	Pass	::test_higher_discount_rate_lowe
Source check on load-bearing input	Wrong number copied from filing	XBRL cross-check of stored historicals	176/176 match	Pass	data_ingestion.verify_against_xb MODEL_SUMMARY

Claim/output	Failure mode	Test and expected result	Actual	Disposition	Evidence
Independent source check	n/a	Checked 2026 capex guidance in the 10-Q Q2 on page 30: " We are focused on long-term growth opportunities through critical, high-value investments. We currently expect our capital expenditures to be in excess of $25 billion in 2026, driven by our AI initiatives, including investments in compute infrastructure and data centers, the expansion and ramp of our manufacturing and R&D production lines and facilities, and growth in our fleet of company-operated AI-enabled assets and our retail, service and charging footprint." The model uses $25,000 million (25 billion)which is a match to Tesla's statement	Match: model 2026 capex = $25,000M	Pass	10-Q Q2 2026 p. 30; tsla_model/assumptions.py (capex)

Claim/output	Failure mode	Test and expected result	Actual	Disposition	Evidence
Peer-policy consistency	Peers defined after seeing results; mixed multiple definitions	Peer policy stated before selection; EV and equity multiples on same basis	Policy written in Lab 08 before multiples were computed, carried into `docs/peer-policy.md`. Check on 2026-10-19 found the document defined EV = market cap + debt − cash, but the code (`valuation.py multiples_crosscheck`) uses market cap + debt; the document was corrected to match the code and the missing cash deduction is stated as a limitation	Pass after correction	`docs/peer-policy.md`; `tsla_model/valuation.py`
Cold run	README doesn't work for a stranger	Fresh clone from GitHub into a new folder, new virtual environment, install requirements, run tests and export	2026-10-19, own run, 21:39–21:45 EDT (6 minutes): clone, install, **71 passed**, export printed FCFE 29.05 / FCFF 27.22, status OK/OK. First issues: (1) `cd $HOME\Desktop` failed (OneDrive Desktop) — used Downloads; (2) the pushed README still said "66 passed" and plain `streamlit run`, which is not recognized on this PC — fixed to `python -m streamlit run` (README update committed after the run)	Pass	Terminal output from the run; `README.md`

Claim/output	Failure mode	Test and expected result	Actual	Disposition	Evidence
Highest-risk claim	The model's value is far below the price mainly because of the capex plan and operating drivers, not an error	Locked Changed-Input Record (Model 3/Y deliveries −10%, prediction committed first) plus FCFF/FCFE cross-check and reverse DCF	Value fell 10.6% as predicted (~10%); decision unchanged; FCFF $27.22 vs FCFE $29.05 reconcile (gap = debt and NCI in the bridge); reverse DCF needs ~$451B robotaxi revenue by 2030	Pass	Locked record below; output/visible_output.md

Known oddity to explain: terminal value is **110%** of DCF value in base (and negative in bear and recession). This happens because FY2026–27 FCFE is negative, so the PV of the forecast years is below zero. Explain it in Video 2 rather than hiding it.

Requirement gap resolved (2026-10-19): Project 1 requirement 4 asks for a **FCFF DCF with WACC and an enterprise-to-equity bridge** driven by the pro-forma. Codex added tsla_model/fcff.py: FCFF from the pro-forma's EBIT, D&A, capex and working capital, discounted at a WACC of 10.33%, with a 30-Jun-2026 enterprise-to-equity bridge (FCFF $27.22/share vs. FCFE $29.05/share). New validation rows:

Claim/output	Failure mode	Test and expected result	Actual	Disposition	Evidence
FCFF known answer	Formula error	Three-year synthetic case with round numbers, checkable by hand	Matches	Pass	tests/test_fcff.py::test_fcff_three_
FCFF boundary	g ≥ WACC gives nonsense	Terminal growth at or above WACC → refuse	Refuses	Pass	::test_fcff_refuses_terminal_grow
FCFF monotonicity	Value moves the wrong way	Higher WACC → lower value; higher growth → higher value	As expected	Pass	::test_fcff_wacc_and_growth_mo
Enterprise-to-equity bridge	Bridge lines don't add up	Bridge lines sum to equity value and per-share value	Ties	Pass	::test_fcff_q2_bridge_ties_to_equ
FCFF refuses when broken	Broken statements still valued	A broken balance sheet blocks the FCFF valuation	Refuses	Pass	::test_broken_balance_sheet_blo
FCFF vs FCFE reconciliation	Hidden convention mismatch	FCFE = FCFF − after-tax interest + after-tax interest income + net borrowing, residual shown not forced	Residual −$60M/yr = NCI share of income	Pass, explained	output/visible_output.md (FCFF-to-FCFE reconciliation)

Locked Changed-Input Record (human prediction first, AI closed)

> The Lab 11 record (`Lab_11_proforma_sensitivity.md`, locked 2026-09-29 13:40) was run on the simplified Lab 10 model, not on the app of record. Its first commit (`241f8b1`, 13:49) already contained both the prediction and the results, so there is **no commit of the prediction made before the run**. Use it as supporting evidence, and do a fresh record on the app:
>
> 1. With AI closed, write your prediction in the table below (input, old → new, expected direction, rough size, decision effect).
> 2. **Commit and push it.** That commit hash is your proof that the prediction came before the run.
> 3. Make the change in the app, record before/after, then fill in the second table.

Precommit

Commit: 3bbe9b5 (pushed 10:41 AM)

10/19/2026 10:34 am Model 3/Y deliveries, all forecast years, Tesla sells 1,650k–1,900k cars per year I want to lower this by 10% every year, my expected direction is that Tesla's value per share goes down by about 10% because their sales decrease by 10%, if fewer cars are sold that leads to less revenue which leads to less profit, less cash and a lower value. I still think that my decision will be to defer Tesla if they're selling 10% less cars a year.

Result

Before I made the change the value was $29.48 per share, after dropping the amount of cars sold by Tesla in a year, the value went down to $26.36 per share. This model ran on live data from today's Tesla market price of $367.81. Overall, the value drop was 10.6% decrease which is really close to my prediction, and my choice to defer stays the same. Tesla's revenue only dropped by 4.9% since they generate revenue from other products and services, but the value dropped by 10.6% because of Tesla's fixed costs such as the factory spending and depreciation that happen even if their sales decreases.

Named AI-use register

Tool/model	Date	Task	Output used	Independent check	Disposition	Effect
ChatGPT / Codex	2026-09-03	Helped draft the Sept 3 research report (my Edition A)	Report text	Sources checked against the 10-K	Accepted; disclosed as AI-assisted Edition A (docs/research-evolution.md)	Baseline position

Tool/model	Date	Task	Output used	Independent check	Disposition	Effect
Claude Code, claude-opus-5-5	2026-09-24	Built the original tsla-proforma-model/ (statements, checks, FCFE valuation, app, 66 tests) and Lab 09/10 files	Model of record	Tests; ABG known answer $291.75; 176 XBRL matches	Accepted after tests	Core model
Codex, GPT-5	2026-10-08	Requirement review, AGENTS.md, README decision frame, peer policy, normalization and challenge records	Docs	Read against Lab 08 and filings	See docs/ai-use-log.md	Documentation
Codex, GPT-5	2026-10-19	Built FCFF/WACC module and bridge; README and peer-policy fixes	fcff.py, test_fcff.py, docs	Tests run independently: 11 failed, then 71 passed after fix	Corrected, then accepted (see next row)	FCFF $27.22
Claude Code, claude-opus-5-5	2026-10-19	Planning, review of Codex output, drafts of the decision memo, Edition B and video scripts	Drafts	Every number checked against visible_output.md; I rewrote the memo's recommendation and triggers and Edition B's conclusion	Modified	Written deliverables
Claude (Cowork), model id claude-opus-5-5	2026-10-06	Read the course repo and this repo; listed the Project 1 gaps; ran the cold-run tests; wrote export_visible_output.py, the root README project map, and this draft record	Visible output files, README, draft tables	Tests rerun (66 pass); base value $29.05 matches MODEL_SUMMARY ($29.03 at a slightly different price); ABG known answer rerun ($291.75)	Accepted, then modified on 2026-10-19 (README cold-run commands updated; this record completed)	Packaging only; no change to valuation

Tool/model	Date	Task	Output used	Independent check	Disposition	Effect
Corrected AI output: Codex FCFF build	2026-10-19	Codex's first FCFF build read the cost of debt as a single parameter (a.p("interest_rate_debt")) instead of the yearly driver	Rejected first version	Ran the tests independently: 11 of 71 failed and the export crashed with KeyError: 'interest_rate_debt'; after Codex's fix, Codex said it could not run tests, so I re-ran them: 71 passed, export FCFF $27.22	**Corrected**, then accepted	Shown in Video 1

Limitations, monitoring, and reversal triggers

My model's biggest limitation is that robotaxi revenue is my own guess, because Tesla doesn't yet report it, and the stock price really depends almost entirely on it. Also, most of my value comes from after 2030, which is a long time in the future, because Tesla's heavy spending makes the next two years negative. So small changes in my long-term guesses move the answer a lot. Tesla also doesn't report prices for each car model, so I had to use an estimate for those.

I'll watch for three things: Tesla reporting robotaxi revenue on its own line, Tesla cutting its spending plans, and car sales dropping below 1.5 million a year. Any of those could make me rerun the model and could change my recommendation.

Supporting numbers: the market price needs about **$451B of robotaxi revenue by FY2030** (vs. $8B in the base case), or a cost of equity of about **3.6%**, or perpetual growth of about **9.8%** (tsla-proforma-model/output/visible_output.md).